

Table A.4: Tool argument summaries for Agentic mode.

Tool	Key arguments
notebook_list_cells	url (notebook session URL)
notebook_get_cell	url, cell_index (1-based)
insert_cell	url, tab_id, position, cell_type, source
edit_cell_by_index	url, tab_id, cell_index, source
delete_by_index	url, tab_id, cell_index
run_cell	url, tab_id, cell_index

A.8 Host Configuration Flags

Table A.5 lists environment variables governing Agentic behaviour during benchmarks.

Table A.5: Agentic configuration flags (config.py).

Variable	Default	Effect
LLM_AGENTIC_ENABLED	on	Enables Agentic tool schemas
AGENTIC_MANDATORY_TWO_PHASE	on	Round 0 read-only; round 1 implement
AGENTIC_MAX_TOOL_ROUNDS	2	Hard cap on LLM API calls per message
AGENTIC_FIRE_AND_FORGET	on	Dispatch batch without per-cell replanning
CEREBRAS_MODEL	zai-gpt-4o	LLM 4.7 model identifier

A.9 Reproduction Commands

From the repository root, with CEREBRAS_API_KEY set in testing/host/.env:

```
python testing/host/scripts/run_agent_tests.py --live-llm
python testing/host/scripts/run_ask_tests.py --live-llm
python testing/host/scripts/run_code_tests.py --live-llm
python testing/host/scripts/generate_fyp_dissertation_report.py
```

Log outputs appear under testing/host/data/logs/, including per-test JSON results and trace files (agentic_tool_trace.jsonl). The consolidated report is written to FYP_DISSERTATION_BENCHMARK_REPORT.md.